

[Figure 11 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 1** (homogeneous, target SM = 3.00 MPa), with **1 % noise** added to the measured surface displacements. Three rows correspond to 3, 6, and 9 displacement datasets.

Explicit (left column): (a) 3 fields: SM $\in [3.0013, 3.0013]$ MPa, uniform red; (b) 6 fields: SM $\in [3.0000, 3.0001]$ MPa, visible green/yellow patches emerge (non-monotone: noisier than (a) or (c)); (c) 9 fields: SM $\in [2.9997, 2.9997]$ MPa, uniform red.

Implicit (right column): (d) 3 fields: SM $\in [2.9981, 3.0083]$ MPa, visible blue patch along the left edge; (e) 6 fields: SM $\in [2.9978, 3.0006]$ MPa, small central blue blob; (f) 9 fields: SM $\in [2.9995, 2.9998]$ MPa, small blue patches remain near the edges.

Takeaway: under 1 % noise, both methods stay quantitatively close to the true 3.00 MPa value (≤ 0.01 MPa deviation). The explicit method’s 6-field result is the noisiest explicit panel on this page — an artifact not discussed in the surrounding prose.

Fig. 11. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (1 % noise is introduced into the measured data, Unit: MPa).

Table 1

Comparison of the average relative error in the shear modulus distribution obtained by implicit inverse and explicit inverse methods for the layered structure.

Inverse method		Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets		3	6	9	3	6	9
1 % noise	Sample 1	0.04 %	0.00 %	0.01 %	0.18 %	0.01 %	0.01 %
	Sample 2	11.56 %	6.40 %	4.53 %	13.20 %	7.44 %	4.69 %
	Sample 3	14.03 %	6.84 %	6.71 %	14.55 %	11.37 %	8.58 %
5 % noise	Sample 1	0.36 %	0.22 %	0.003 %	0.11 %	0.15 %	0.003 %
	Sample 2	15.29 %	11.24 %	8.12 %	24.44 %	11.96 %	8.61 %
	Sample 3	25.00 %	9.20 %	8.14 %	19.03 %	13.61 %	10.58 %

with a 1 % noise level. Furthermore, we provided the values of the regularization factors for each case depicted in Figs. 8–16 in Table 2.

Case 2: A layered ring structure

In the second numerical example, we focused on a layered ring structure, as illustrated in Fig. 17. To discretize the problem domain, we employed 7610 triangular elements. The deformation of the ring structure was induced by applying inner pressures ranging from 10 mmHg to 50 mmHg, and we were only able to “measure” the displacement on the inner wall. The shear modulus values of the inner and outer layers were 0.129 MPa and 0.386 MPa [37], respectively. Our approach for the inverse problem relied on utilizing the measured displacement on the inner wall to determine the shear modulus distribution. Initially, we provided an initial guess of the shear modulus distribution,

as shown in Fig. 18(a), where four layers were predefined. The shear modulus values, from the outermost layer to the innermost layer, ranged from 0.01 MPa to 0.04 MPa. During the inverse problem-solving process, we optimized both the shape of the interface between neighboring layers and the shear modulus value of each layer. In the initial assumption, the ring structure is considered to consist of 4 layers. For the explicit inverse method, each interface is represented by 14 control points, leading to a total of 56 optimization variables, which includes 4 shear moduli. On the other hand, the implicit inverse method involves a much larger number of optimization variables, specifically 4030 variables in total. Fig. 18 displays the variation of the reconstructed shear modulus distribution using the explicit inverse method for the noise free case. Notably, the reconstruction achieved through the explicit method closely resembled the ground truth. However, the same cannot be said for the implicit inverse method, where the reconstructed shear modulus distribution performed significantly worse than the explicit inverse method even for the noise free data (see Fig. 19(d)-(f)).